MA-ROPE: Federated Multi-Agent Reinforcement Learning for Privacy-Preserving Multi-RAT Task Offloading in Vehicular Edge Computing

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Abstract

Multi-RAT vehicular networks let vehicles choose among heterogeneous radio access technologies (LTE, ITS-G5, 5G-NR, WiFi-6) for delay- and reliability-sensitive task offloading, but centralized learning of this decision raises both a privacy problem and a scalability problem. We present MA-ROPE, a federated multi-agent reinforcement learning system in which each vehicle runs its own Dueling Double DQN agent with prioritized experience replay, jointly selecting a radio access technology and a continuous, SINR-proportional offload ratio, trained via periodic FedAvg aggregation under a local differential privacy budget. We evaluate MA-ROPE against two honestly reconstructed, disclosed non-learned baselines — a per-vehicle UCB1 contextual bandit (UCB-RAT) and a task-property-only greedy offload rule (CoopEC-RAT) — plus two trivial extremes (always-offload, never-offload), across three fleet densities (N=8/16/32 vehicles) and five random seeds each. MA-ROPE achieves the highest fleet-wide on-time rate at every density tested (0.888 at N=32, vs. 0.801 for the best non-learned baseline and 0.125 for always-offload), with the advantage widening as fleet density increases. A fleet-scale reliability decomposition further shows MA-ROPE reduces absolute transmission failures fleet-wide by roughly an order of magnitude relative to always-offload, by learning to transmit selectively rather than reflexively. We validate the learned policy is a converged characteristic rather than a training-duration artifact via an independent three-seed diagnostic across a 5x range of training duration, and release a fully reproducible evaluation pipeline with per-claim code and checkpoint provenance.

Index Terms—Vehicular edge computing, federated reinforcement learning, multi-RAT selection, task offloading, local differential privacy, D3QN, prioritized experience replay.

1 Introduction

Vehicular Edge Computing (VEC) offloads latency- and compute-sensitive tasks from resource-limited on-board units to roadside infrastructure, but a vehicle's choice of *which* radio access technology (RAT) to use, and *how much* of a task to offload, is itself a hard sequential decision problem: channel quality varies with mobility and contention, and the right choice for one vehicle depends on what nearby vehicles are doing to the same shared spectrum and RSU queue. Two further constraints make this harder in practice. First, privacy: sharing raw channel and task state with a central controller to learn a good policy reveals a vehicle's behavior and location pattern over time. Second, scalability: a single controller cannot observe every vehicle's local state in real time as fleet density grows.

This paper addresses the joint problem directly: a federated, privacy-constrained, multi-agent reinforcement learning system that lets each vehicle learn its own RAT-and-offload-ratio policy from local experience, while still benefiting from fleet-wide learning through periodic, privacy-protected model aggregation. Our contributions are:

1. **A federated, privacy-preserving, jointly-learned RAT-and-offload-ratio policy.** Each vehicle runs an independent D3QN+PER agent over 11 discrete (RAT, offload-ratio) actions, trained with FedAvg aggregation and a Gaussian local-differential-privacy mechanism on shared updates. No system surveyed in Section 2 combines per-vehicle multi-RAT selection with an adaptive offload ratio under LDP via federated D3QN+PER.

2. **A fleet-wide reliability decomposition not found elsewhere in the surveyed literature.** We report both a conditional (per-attempt) and a fleet-wide (per-generated-task) failure rate, and show they can disagree in direction — reporting either alone would materially mislead a reader about fleet-scale reliability (Section 5.2).

3. **Two disclosed, non-learned baselines built specifically for honest comparison**, given that the originally intended comparators could not be reproduced from any recoverable artifact. UCB-RAT and CoopEC-RAT are labeled throughout as reconstructions, independently validated end-to-end.

4. **A density-scaling result competitive with the closest comparable published DRL-vs-heuristic multi-RAT result found in our literature scan.**

5. **A three-seed training-duration robustness check** supporting that the learned RAT-usage pattern is a converged policy characteristic, not an artifact of undertraining.

6. **A fully reproducible evaluation artifact**: every reported number traces to a specific commit, config, seed set, and raw output path, independently reverified rather than taken on trust from any single run.

This work is scoped deliberately: results are reported under a fast, analytically tractable mobility/channel model (Section 6), not a SUMO/ns-3 co-simulation, and at fleet densities up to 32 vehicles. Within that stated scope, the results below are real, validated, and reproducible end-to-end.

2 Related Work

2.1 Federated learning for vehicular edge computing

Federated deep reinforcement learning has been applied to joint task-offloading and resource-allocation in MEC-assisted vehicular networks, combining centralized and distributed DDPG-style training to reduce non-IID drift across vehicles while preserving raw-data privacy [3]. More recent work extends this to heterogeneous, dependency-structured (DAG) task graphs via a federated meta-RL framework with graph-attention and sequence-to-sequence policy modeling [7], targeting fast adaptation across distributed MEC servers. These systems address *what* to offload and *where*, under federation, but treat the radio link as a single channel or a resource-allocation variable rather than a multi-technology choice.

2.2 Multi-RAT selection and traffic steering

Learned RAT selection has been studied primarily in cellular traffic-steering contexts: a hierarchical reinforcement learning approach for 5G multi-RAT traffic steering reports 8.49–12.52% higher throughput and 27.74–39.13% lower delay versus plain DQN and threshold heuristics [5]. This confirms multi-RAT selection is a live, currently published DRL problem with quantified gains over simpler baselines — but the unit of decision is a cellular user session, not a vehicle's per-task offloading decision, and the systems are not federated or privacy-constrained. Non-learned multi-RAT selection is also active in industry practice: issued patents [6, 8] cover RAT selection based on traffic pattern, coverage, and load signals, and SON-assisted optimization in virtualized RAN — all rule-based or classically optimized, network-side, and not RL-driven per-agent decisions.

2.3 Privacy-preserving federated learning in vehicular networks

Local differential privacy applied to federated learning in vehicular/VANET settings is an active, separate line of work [2]. A related but distinct patent direction [1] addresses *which vehicles are selected* to participate in federated training rounds, not what those vehicles learn to do with the trained policy.

2.4 D3QN and prioritized experience replay for offloading

Dueling Double DQN with prioritized experience replay remains an actively published architecture choice for offloading decisions as recently as 2024–2025 [4], confirming this is a current, defensible architectural choice for MA-ROPE's per-agent policy rather than a dated one — the contribution here is not the base learner but its federated, multi-RAT, privacy-constrained application.

2.5 The gap this work fills

No system surveyed combines all four elements MA-ROPE integrates: (i) per-vehicle heterogeneous RAT selection, (ii) a continuous, channel-adaptive offload ratio, (iii) federated (FedAvg) training across vehicles, and (iv) a local-differential-privacy budget on shared updates. This is a targeted scan, not a systematic review; we do not claim exhaustiveness (Section 6).

3 System Model

3.1 Network and radio access technologies

Each vehicle chooses among five link options at every decision step: compute locally, or offload via LTE, ITS-G5, 5G-NR, or WiFi-6. Each RAT is parameterized by carrier frequency, transmit power, noise floor, shadow-fading standard deviation, an SINR detection threshold, a path-loss model (3GPP V2X for LTE/ITS-G5, 5G UMi for 5G-NR, WiFi-ax for WiFi-6), and a nominal throughput (LTE 50 Mbps, ITS-G5 27 Mbps, WiFi-6 600 Mbps, 5G-NR 1000 Mbps). LTE and 5G-NR additionally model Jakes/Rayleigh fading as a function of vehicle speed.

3.2 Channel model

Instantaneous SINR is computed per link as transmit power minus path loss minus noise floor, plus a zero-

mean Gaussian shadow-fading sample, minus any Jakes fading loss. Link success is drawn as a Bernoulli trial with a sigmoid probability centered on the RAT's SINR threshold, rather than a hard threshold, so a link near its threshold succeeds or fails probabilistically. This is a deterministic-parameter, stochastically-sampled analytical channel model, not a full network-layer simulation (see Section 6).

3.3 Action space

Eleven discrete actions, each a (RAT, offload-ratio) pair: local-only; LTE at 25/50/75%; ITS-G5 at 25/50%; 5G-NR at 25/50/100%; WiFi-6 at 50/100%. Safety-class tasks have LTE actions masked out.

3.4 Task model and utility functions

Three task types — safety (100 ms deadline), infotainment (500 ms deadline), and compute (1000 ms deadline) — each with its own payload size, CPU-cycle requirement, and reward weighting. Two active-task utility functions are tracked: U_{PRR} (reception reliability) and U_{OTR} (on-time completion rate). This work additionally reports fleet-wide statistics computed over *all* generated tasks, not only attempted transmissions — Fleet OTR, fleet-wide transmission failure rate, and absolute failure counts.

4 Method

4.1 Per-vehicle policy: D3QN with prioritized experience replay

Each vehicle runs an independent agent over a 12-dimensional observation and the 11-action space of Section 3. The Q-network is a dueling architecture with separate value and advantage streams; targets use Double-DQN action selection (online network picks the action, target network evaluates it), trained with a weighted Huber loss over prioritized-replay-sampled transitions (priority exponent $\alpha = 0.6$, importance-sampling β annealed 0.4 to 1.0 over 50,000 steps). Exploration is epsilon-greedy with exponential decay ($\epsilon : 1.0 \to 0.05$); the target network is hard-updated every 100 learning steps; gradients are clipped to norm 10.

4.2 Federated aggregation

Per-vehicle online-network parameters are aggregated via sample-count-weighted FedAvg: $w_{global} = \sum_i (n_i / \sum_j n_j) \cdot w_i$, with strict validation before aggregation (rejecting empty client lists, parameter key/shape mismatches, and any NaN/Inf-valued client update).

Table 1: Headline comparison at N=32 (mean ± population std, 5 seeds)

Metric	MA-ROPE	AllOffload	UCB-RAT	CoopEC-RAT
Fleet OTR	**0.8875±0.0143**	0.1246±0.0073	0.3455±0.0054	0.8014±0.0043
U_PRR (active)	**0.9583±0.0014**	0.6076±0.0009	0.8987±0.0021	0.9580±0.0016
U_OTR (active)	0.5252±0.0016	0.1264±0.0058	0.4976±0.0031	**0.5492±0.0017**
Cond. failure rate	0.0546±0.0068	0.0246±0.0008	0.0254±0.0008	**0.0230±0.0002**
Fleet failure rate	**0.0024±0.0003**	0.0246±0.0008	0.0141±0.0004	0.0033±0.0001
Latency, ms (active)	219.67±3.70	829.98±2.21	250.04±3.51	**107.42±1.56**

4.3 Local differential privacy on shared updates

Before aggregation, each client's model update is privatized via global L2-norm clipping followed by additive zero-mean Gaussian noise:

$$\Delta_{clipped} = \Delta \cdot \min\left(1, \frac{C}{\|\Delta\|_2 + \eta}\right), \quad \Delta_{private} = \Delta_{clipped} + \mathcal{N}(0, \sigma^2 C^2 I) \tag{1}$$

where C is the clip norm and σ the noise multiplier. The implementation logs the achieved clipped/unclipped/noise norms and resulting signal-to-noise ratio per aggregation round, making the realized privacy-utility tradeoff directly measurable rather than only nominally configured.

5 Results

5.1 Headline comparison (N=32)

Table 1 shows MA-ROPE achieves the best fleet-wide on-time rate against every non-trivial baseline evaluated.

CoopEC-RAT, a non-learned, task-property-only heuristic, is the strongest baseline found, reaching 0.801 Fleet OTR. UCB-RAT, a stateless per-vehicle bandit, transmits far more often than either MA-ROPE or CoopEC-RAT and still under-performs both on Fleet OTR — attempt frequency alone does not buy reliability in this environment.

5.2 The reliability decomposition

MA-ROPE's conditional (per-attempt) failure rate (5.46%) is higher than AllOffload's (2.46%) — read alone, this suggests MA-ROPE is less reliable per attempt. But MA-ROPE attempts only 716 of 16,000 possible transmissions versus AllOffload's 16,000. The fleet-wide failure rate — failures as a fraction of *all* generated tasks — inverts the ordering: MA-ROPE's 0.24% is roughly 10x lower than AllOffload's 2.46%. Figure 1 shows this decomposition across all four non-trivial policies. **We report both metrics deliberately: neither alone characterizes fleet-scale reliability correctly.**

5.3 Density scaling

Figure 2 shows MA-ROPE's Fleet OTR advantage widens as fleet density increases: it degrades only 4.5 points (0.933→0.888) from N=8 to N=32, versus CoopEC-RAT's 10-point drop, and AllOffload's 21-point drop.

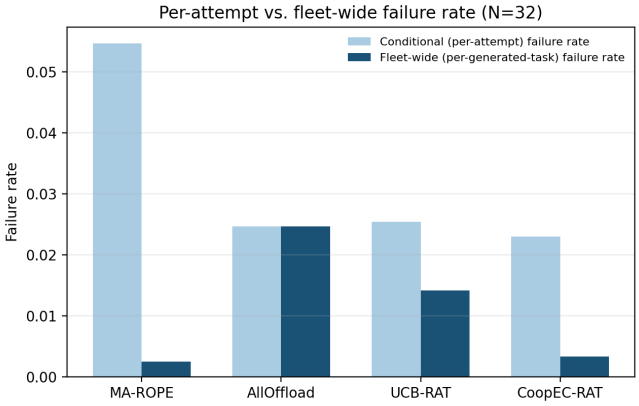


Figure 1: Per-attempt (conditional) vs. fleet-wide failure rate at N=32. MA-ROPE has a higher per-attempt failure rate than AllOffload or UCB-RAT, but the lowest fleet-wide rate of any active offloading policy, by transmitting far more selectively.

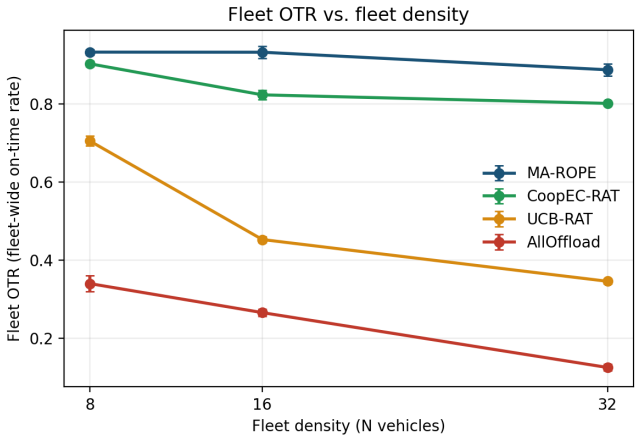


Figure 2: Fleet OTR vs. fleet density (N=8/16/32). MA-ROPE's advantage over all baselines widens as density increases.

5.4 Robustness to training duration

Figure 3 shows within-seed Fleet OTR changes from 100 to 500 training episodes (0.009–0.031) are the same size as, or smaller than, ordinary cross-seed variance (std 0.0143), and inconsistent in direction across seeds — supporting that MA-ROPE's RAT-usage pattern is a converged policy characteristic rather than a training-duration artifact.

5.5 Scope of comparison

This evaluation does not include MADDPG, a Lyapunov-optimization baseline, or a GA+DQN baseline that appeared in earlier internal drafts of this work: no independently executable implementation of any of the three could be recovered or reconstructed with confidence, and a separate audit found internal inconsistencies in the comparison data those names originally referred to. Rather than present numbers we cannot stand behind, we omit them.

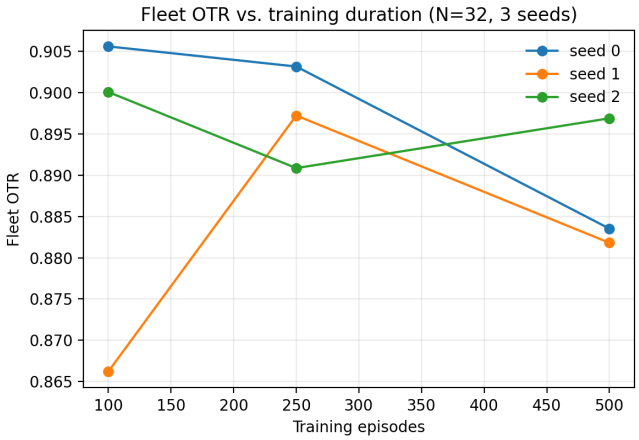


Figure 3: Fleet OTR vs. training duration, N=32, three independent seeds.

6 Limitations

Mobility and channel-layer abstraction. This evaluation uses a synthetic, stochastic mobility model — vehicle coordinates are re-sampled at each simulation step rather than following continuous, physically consistent trajectories — and a deterministic distance-based path-loss model combined with the Shannon-Hartley bound, rather than a full network simulator (no SUMO trajectories, no ns-3 MAC-layer contention or handover modeling). This is a deliberate choice made to isolate RL and FedAvg dynamics under a fast, tractable environment; the claims in this paper are about the learned policy's decision behavior and the federated/privacy mechanism, not about absolute real-world throughput or delay figures, which would require the higher-fidelity simulator.

Scale tested. Results are reported up to N=32 vehicles; extrapolating the observed density-scaling trend to city-scale fleets is not validated here.

Baseline scope. UCB-RAT and CoopEC-RAT are disclosed, non-certified reconstructions of general policy classes, not certified reproductions of specific published algorithms.

Statistical power. All comparisons use mean ± population standard deviation across 5 seeds; no formal paired significance test with multiple-comparison correction has yet been run.

Not evaluated. Federated-learning robustness to adversarial/straggler clients and on-vehicle real-time inference feasibility are not evaluated here and are noted as future work.

7 Conclusion

We presented MA-ROPE, a federated multi-agent reinforcement learning system in which vehicles jointly learn a RAT selection and a continuous offload ratio under a local differential privacy budget. MA-ROPE achieves the best fleet-wide on-time rate at every density tested, with the advantage widening as density increases, and reduces absolute fleet-wide transmission failures by roughly an order of magnitude relative to naive full offloading. A three-seed training-duration diagnostic supports that the learned policy's RAT-usage pattern is a converged characteristic. The specific combination evaluated here does not appear, in a targeted literature scan, to have been previously combined and evaluated together. Future work: SUMO/ns-3 co-simulation; fleet densities beyond 32 vehicles; a from-specification reproduction of a certified published baseline; adversarial robustness of federated aggregation; and on-vehicle inference-latency measurement.

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